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SARDAR PATEL UNIVERSITY

B.C.A. – III SEMESTER EXAMINATION 2023 (OLD COURSE)

US0CBCA03: ADVANCED DATA AND FILE STRUCTURE

Date: 27/10/2023 (Friday)

Time: 02.00 P.M. TO 05.00 P.M.

TOTAL MARKS: 70

NOTE: FIGURE TO RIGTH INDICATES FULL MARKS OF THE QUESTIONS.

Q-1 Select the appropriate option for the following.

10

- 1 _____ is a sequence of consecutive edged from the source node to destination node.
A. Path
B. Leaf
C. Edge
D. Height
- 2 An array is a _____ data structure.
A. Unordered
B. Non-composite
C. Linear
D. Heterogeneous
- 3 The number of elements in an array is called the _____ of the array.
A. Size
B. Base
C. Type
D. Index
- 4 A digraph is known as _____ graph.
A. Directed graph
B. Multiple graph
C. Double graph
D. Undirected graph
- 5 A binary tree has at most _____ child.
A. Two
B. Three
C. One
D. None of these
- 6 There are at most _____ passes required to perform bubble sort.
A. $N - 1$
B. N
C. $N + 1$
D. $N / 2$
- 7 The process of finding the data from its data structure is called _____.
A. Searching
B. Detection
C. Sorting
D. None of these
- 8 All types of external storage device support a _____ file organization.
A. Direct
B. Indexed Sequential
C. Sequential
D. None of these
- 9 CDC stands for _____.
A. Control Data Cycle
B. Control Data Corporation
C. Control Data Centre
D. None of these
- 10 A collection of records involving a set of entities with certain aspects in common and organized for some particular purpose is called as a _____.
A. Key
B. File
C. Record
D. None of these

P.T.O.

Q-2	Give Short answers of the following questions (ANY TEN).	20
1	List 2 applications of an array.	
2	Define : Node and Branch	
3	Draw the Binary Tree for $(A-B)+c*(E/F)$	
4	Define Directed and Undirected Graph.	
5	Define bubble sort.	
6	List the applications of sorting.	
7	Define: File, Database	
8	Write down the syntax and purpose of open statement for Input mode.	
9	List any 2 applications of tree.	
10	What is connected Graph?	
11	Difference between Searching and sorting.	
12	What do you mean by file organization? Why it is Required?	
Q-3	(A) Explain 2-D array with declaration and initialization.	05
	(B) Explain application of tree.	05
	OR	
Q-3	(A) Define array. Explain 1-D array with declaration and initialization.	05
	(B) Explain application of Array.	05
Q-4	(A) Define Binary tree. Draw a binary tree for $(A+B*C)/(A+B)*C$	05
	(B) Explain lexically ordered tree.	05
	OR	
Q-4	Define Graph. Explain Directed graph, Undirected graph, connected graph, weekly connected graph with an example.	10
Q-5	(A) Write down the algorithm of bubble sort.	05
	(B) Write down the algorithm of Sequential search	05
	OR	
Q-5	(A) What is searching? List and explain searching techniques with algorithms.	05
	(B) What is sorting? List sorting techniques and explain any one of them with algorithm.	05
Q-6	(A) Write a detail note on processing of index sequential file.	05
	(B) Write a detail note on processing of Direct file	05
	OR	
Q-6	(A) Write a detail note on processing of sequential file.	05
	(B) Explain the structure of index sequential file supported by CDC.	05

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